

Performance Testing

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	[e.g., JMeter, Locust, k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	[e.g., Local / Cloud / Hosted Server]
Test Date	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.
2	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.

		response time for agricultural data processing.	response time for agricultural data processing.	
3	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.
4	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.
2	Response Time (Max)	< 5 seconds	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.
3	Throughput (Req/sec)	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.
4	Error Rate	< 1%	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.

5	CPU Utilization	< 80%	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.
6	Memory Utilization	< 80%	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.	Performance testing completed with stable execution, efficient load handling, and optimized response time for agricultural data processing.

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]